

THE UNIVERSITY OF AUCKLAND

SEMESTER TWO, 2023

Campus: City

MECHENG 726

Acoustics for Engineers

Time Allowed: 3 hours and 30 min additional time

NOTE: Answer All Questions

This Exam is worth 60% of your final grade

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I confirm that by completing this exam I agree to the above statements in full.

- 1(a)** An acoustical engineer recorded the sound of the linear swept sine signal (100 to 8000 Hz; 2 seconds long) played from a loudspeaker in an anechoic chamber using a digital sound recorder equipped with a stereo microphone. Figure (A) below shows the spectrogram of the sound played from the loudspeaker. Answer the following questions.

Assume the short-time Fourier transform (STFT) is used to calculate the spectrograms provided in this question.

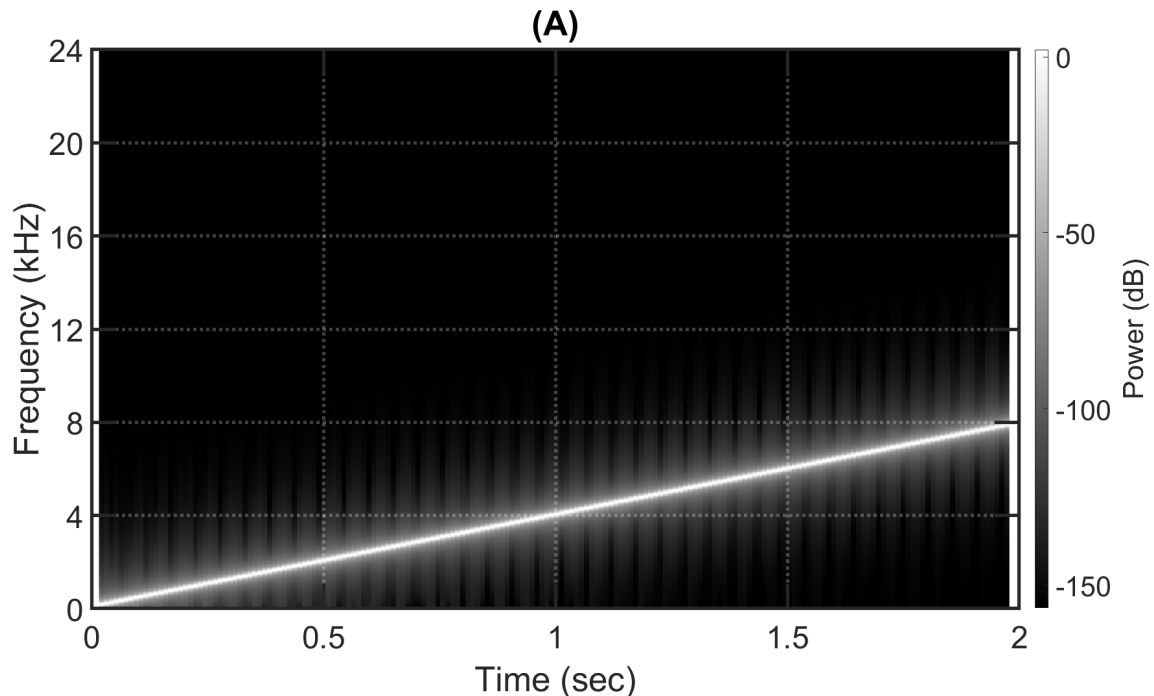


Figure (A): Spectrogram of the sound played from the loudspeaker.

- (i) Choose the correct setting of the digital sound recorder the acoustical engineer should have chosen to record the sound with minimum distortion.

(1 mark)

- ☐ Sampling frequency 8 kHz; Coding type PCM
- ☐ Sampling frequency 48 kHz; Coding type PCM
- ☐ Sampling frequency 8 kHz; Coding type MP3
- ☐ Sampling frequency 48 kHz; Coding type MP3

- (ii) The acoustical engineer attempted to inspect the signal recorded in Q1(i) by their spectrogram, however it ended up with the spectrogram shown in Figure (B) below. What setting should the acoustical engineer change in order to make the spectrogram look similar to that in Figure (A)?

(1 mark)

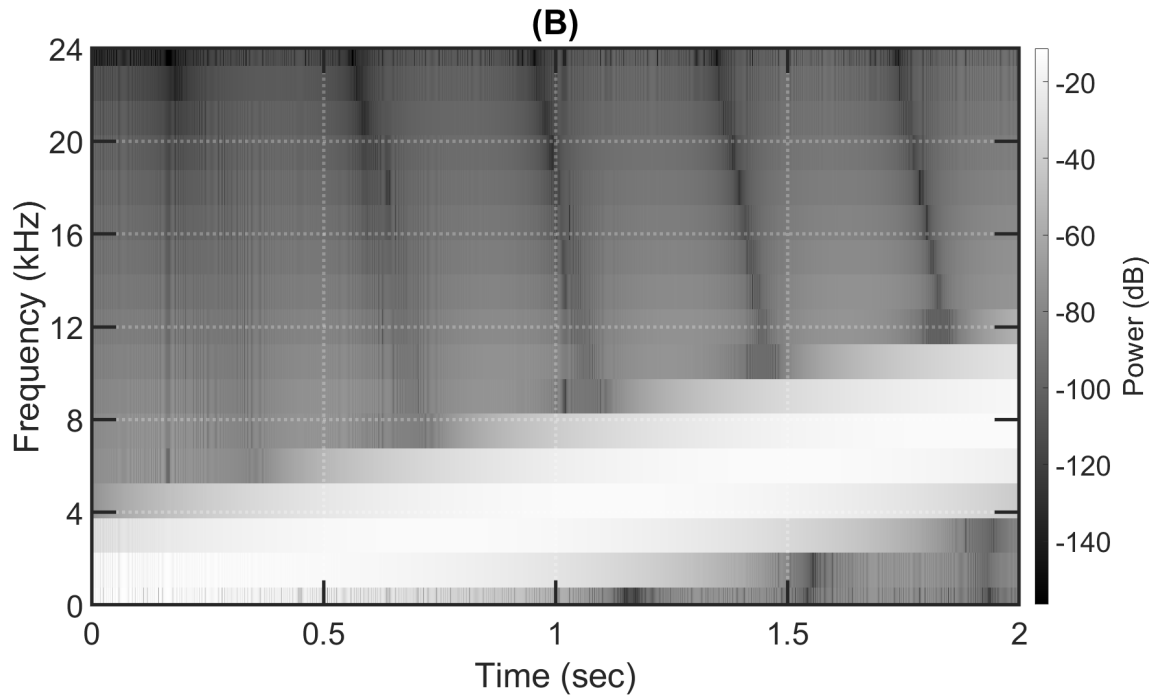


Figure (B): Spectrogram of the recorded audio file.

- ☐ Sampling frequency
- ☐ Window length
- ☐ Bit rate
- ☐ Coding method (PCM or MP3)

(iii) The acoustical engineer applied a digital filter to the linear swept sine signal. The spectrogram of the resultant signal is shown in Figure (C). Which type of filter did the engineer apply?

(1 mark)

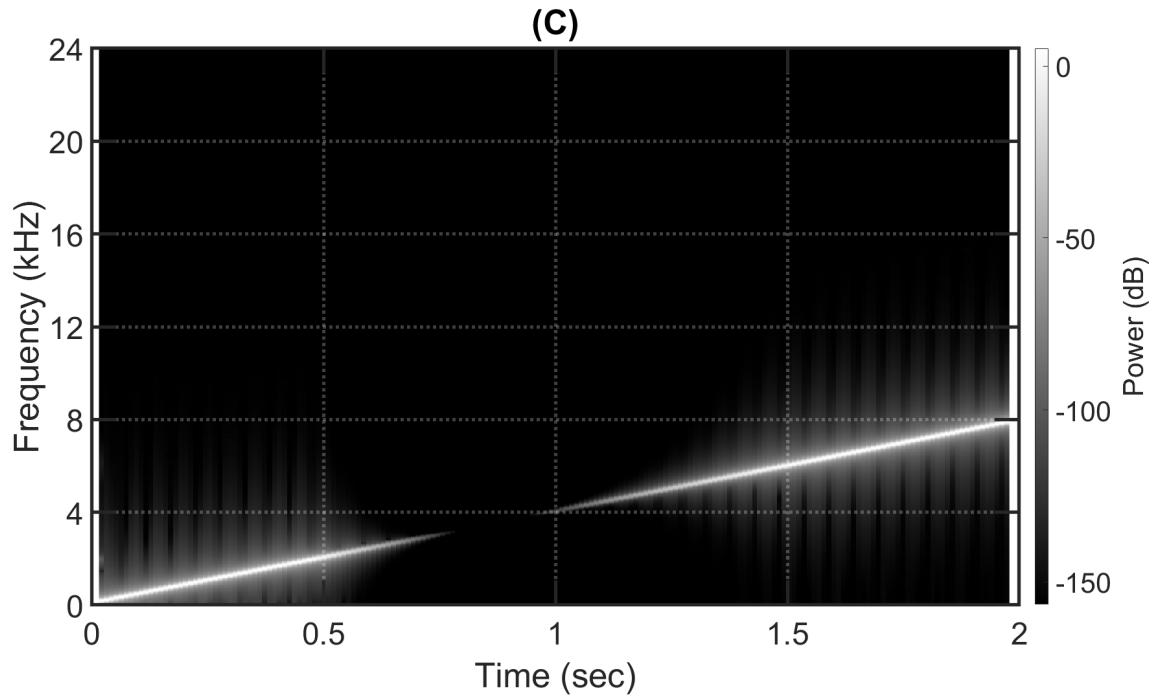


Figure (C): Spectrogram of the filtered swept sine signal.

- ☐ Band limit filter
- ☐ Band pass filter
- ☐ Low pass filter
- ☐ High pass filter

(iv) Assume that the linear swept sine signal is recorded in stereo by pulse code modulation (PCM) with 24 bits/sample and the sampling frequency of the recorder is set at twice the Nyquist rate of the sound. Determine the size of the file saved on the recorder in bytes.
(1 mark)

bytes

Maximum marks: 4

- 1(b)** The following statements are concerned with the physiology of the human auditory perception system. Choose the correct word(s) for each blank to complete the statements.
(2 marks)

i) The cochlea is filled with (particles, air, bones, liquid).

ii) The basilar membrane works as a (high pass, low pass, band pass, band limit) filter.

iii) The (ear drum, Inner hair cells, outer hair cells, pinna, ear canal) controls the movement of the basilar membrane causing an effect known as

(cochlea amplifier, precedence effect, auditory masking, cochlea implant) .

Maximum marks: 2

- 1(c)** You are in Yusuke's lecture seated at the back row of a large lecture theatre as shown in the diagram on the left panel. When you asked a question to Yusuke during the lecture, he did not clearly understand what you said. Discuss ALL potential factors that may have affected Yusuke's understanding of your speech in the given situation and how you could address the issue by referring to the plan and the acoustical characteristics of the lecture theatre given in the diagram.

(4 marks)

Fill in your answer here

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Words: 0

Maximum marks: 4

1(d) One wants to calculate the frequency response of an auditorium by excluding the characteristics of the measurement equipment used. Choose ALL measurement(s) that are required to calculate the required frequency response.

(1 mark)

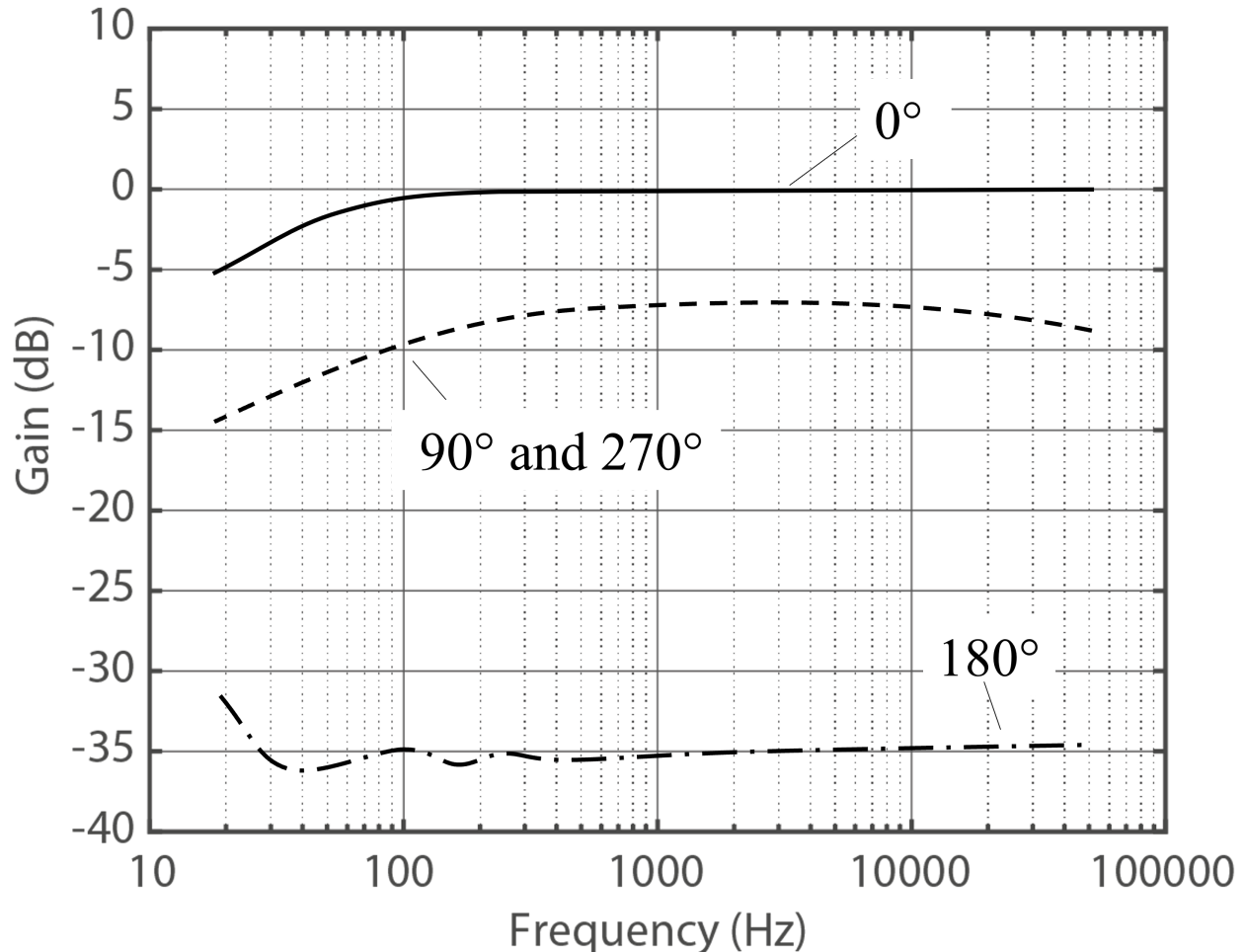
- ☐ Impulse response measured in an anechoic chamber using a calibrated instrumentation microphone and a loudspeaker (model X).
- ☐ Impulse response measured in an anechoic chamber using a calibrated instrumentation microphone and a loudspeaker (model Y).
- ☐ Impulse response measured in the auditorium using a calibrated instrumentation microphone and a loudspeaker (model X).
- ☐ Impulse response measured in the auditorium by using a shotgun microphone and a loudspeaker (model X).
- ☐ Impulse response measured in an anechoic chamber using a shotgun microphone and a loudspeaker (model Y).
- ☐ Impulse response measured in the auditorium using a cardioid microphone and a loudspeaker (model Y).

Maximum marks: 1

- 1(e)** The following diagram shows the frequency response of a microphone measured in an anechoic chamber. The annotations indicate the angle of incidence of the sound relative to the

microphone axis. The directivity of this microphone is (shotgun, uni-directional (cardioid), bi-directional (figure of eight), omni-directional).

(1 mark)



Maximum marks: 1

- 1(f)** One measured the sound pressure level (SPL) of the sound played by a microphone calibrator which plays a 1 kHz tone at 1 Pa using a sound level meter where the reading was 95.5 dBZ. Sound pressure levels measured by this sound level meter should be adjusted by

(multiplying, adding, dividing, subtracting) * dBZ to/from/by the

reading on the sound level meter.

*Note: Your answer must be a positive number.

(1 mark)

Maximum marks: 1

- 1(g)** Imagine you are camping in a dense forest and you can hear the calls of kiwis after dark. Although you cannot see them, you can identify where they are calling from by listening to them using your two ears (i.e. binaural hearing). Explain why you can identify the direction of the kiwis by referring to the mechanism of binaural hearing.
(2 marks)

Fill in your answer here

Format

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x_2

x^2

I_x





















Σ



Words: 0

Maximum marks: 2

1(h) The [room_impulse_response](#) (click the link to download and extract the zip file) has been filtered by a bandpass filter whose passband matches the 1 kHz octave band (from 710 Hz to 1420 Hz). The sampling frequency of the impulse response is 48 kHz. Answer the following questions.

[room_impulse_response](#) (click the link if zip file does not work)

(5 marks)

- Determine the T20 reverberation time of the room for the 1 kHz octave band.
- Describe the process you have used to determine the reverberation time.

Note: Use Matlab (or similar scientific software) for your calculation. Ignore the effect of the characteristics of the microphone and loudspeaker used for the measurement.

Fill in your answer here

Maximum marks: 5

2(a) A worker in a factory is exposed to the following sound levels during their working day:

Time	Level ($L_{A,eq}$)
08:00 – 10:00	80 dB
10:00 – 10:30	50 dB
10:30 – 11:00	90 dB
11:00 – 12:00	80 dB
12:00 – 12:30	50 dB
12:30 – 16:00	80 dB

(i) Calculate the equivalent continuous A-weighted sound pressure level over the 8 hour work day (from 08:00 till 16:00).

(5 marks)

(ii) Based on your answer from part (i), comment on whether the worker should wear hearing protection in order to comply with New Zealand workplace law.

(1 mark)



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Maximum marks: 6

2(b) A helicopter hovering outdoors is modelled acoustically as a source with sound power of 100 Watts and a directivity index $DI = 10\log_{10}(2\cos^2(\theta_0))$, where θ_0 is the angle measured from the vertical.

(i) Ignoring excess attenuation effects, calculate the sound pressure level measured at a location 100 m from the helicopter at $\theta_0 = 45^\circ$.

(4 marks)

(ii) Estimate the sound pressure level measured by a microphone mounted on a large rigid ground board placed on the ground a distance 100 m from the helicopter at $\theta_0 = 45^\circ$. Assume that ground reflections are the only significant excess attenuation effect which needs to be accounted for. State any other assumptions you make and comment on the accuracy of these.

(2 marks)

(iii) Estimate the excess attenuation due to ground reflections if the sound pressure level is measured by a microphone located at a height one metre above the ground at an angle $\theta_0 = 45^\circ$. The ground is modelled as a plane surface with specific acoustic impedance $z = \rho_0 c_0(3 + 3i)$. You may assume that the ground reflected wave adds together incoherently with the incident wave. Assume that the density of air $\rho_0 = 1.2 \text{ kg/m}^3$, and the speed of sound in air $c_0 = 340 \text{ m/s}$.

(6 marks)

(iv) If the measurement is instead made at a location 1 km from the helicopter, describe what factors other than distance, source directivity and ground effects could affect the measured sound pressure level.

(2 marks)



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Maximum marks: 14

2(c) Two rooms are separated by a common wall. The rooms are identical in shape being rectangular prisms with internal dimensions of 4 m long $\times$ 5 m wide $\times$ 3 m high. The common wall measures 5 m $\times$ 3 m. The reverberation time in both rooms is 0.5 s.

(i) Calculate the total sound absorption in each of the rooms.

(2 marks)

(ii) A stereo is playing music in one of the rooms and has a sound power level of 80 dB.

Calculate the reverberant sound pressure level within the room containing the stereo.

(2 marks)

(iii) Calculate the sound reduction index of the common wall required to ensure that the reverberant sound pressure level in the other room due to transmission of the sound from the stereo through the wall is less than 35 dB.

(3 marks)

(iv) An acoustical engineer recommends that the common wall between the two rooms have an STC rating of 37 or greater. Discuss why the STC rating might not be appropriate for quantifying the reduction in noise from the stereo that the wall will provide.

(2 marks)



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Maximum marks: 9

- 2(d)** A starter gun is used to start a running race and produces an impulsive spherical pressure wave with pressure

$$p(r, t) = \frac{P_0}{r} \exp \left\{ - \left(\frac{t-r/c_0}{\tau} \right)^2 \right\},$$

where $P_0 = 100$ Pa.m, $\tau = 10^{-3}$ s and $c_0 = 340$ m/s. Note that r is the distance from the impulsive noise source and t is the time.

(i) Explain why this equation for the acoustic pressure satisfies the three-dimensional acoustic wave equation.

(2 marks)

(ii) Calculate L_{peak} at $r = 1$ m.

(3 marks)

(iii) The person firing the starter gun must comply with New Zealand workplace law. Assuming the person firing the pistol holds it approximately 1 m from their ear, comment on whether the person should wear hearing protection.

(1 mark)

(iv) Estimate $L_{\text{eq}, 60 \text{ s}}$ at $r = 1$ m for the case where the peak pressure occurs in the middle of the 60 second measurement period. Note that for a function of the form

$$f(t) = \exp \left\{ - \left(\frac{t}{\tau} \right)^2 \right\},$$

the mean-square value is given by

$$\overline{f^2} = \frac{1}{2T} \int_{-T}^T \left[\exp \left\{ - \left(\frac{t}{\tau} \right)^2 \right\} \right]^2 dt \approx \frac{\sqrt{2\pi}}{4} \frac{\tau}{T} \text{ for } T \gg \tau.$$

(5 marks)



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Maximum marks: 11

Question 1.c

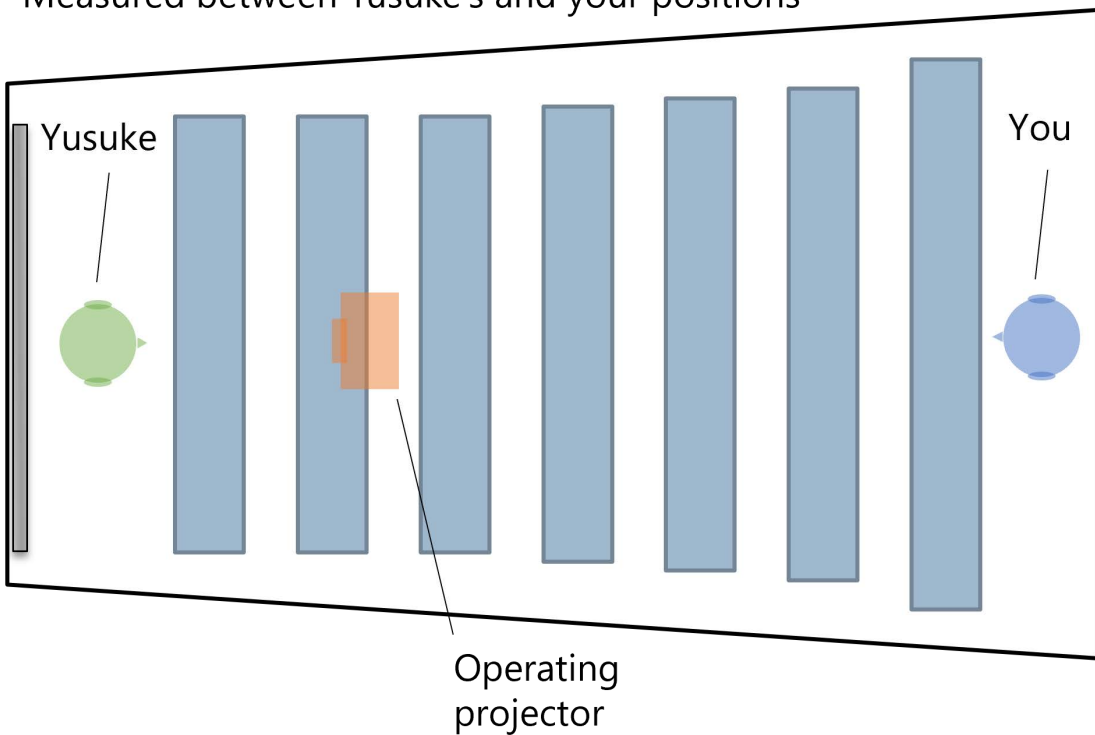
Attached



Reverberation time (T20): 1.2 sec

Clarity (C50)*: 0 dB

*Measured between Yusuke's and your positions



Question 1.g

Attached



